Thiago Amado Costa

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github.com/ThiiagoAC7

Summary

Computer Science Master's Student, currently focusing on studying User Behavior on Social Networks, using Natural Language Processing, Graph Theory, Machine Learning. I can speak Portuguese, English, Intermediate Spanish, basic French.

Skills Python, Javascript, Java, C/C++, SQL, Linux, Machine Learning, Data Science, Deep Learning, NLP

Education

Pontifícia Universidade Católica de Minas Gerais Mar 2025 – Present
Master's in Computer Science

Pontifícia Universidade Católica de Minas Gerais Jul 2019 – Dec 2024
Bachelor's in Computer Science

EPITA - School of Engineering and Computer Science Sep 2023 – Fev 2024
Data Science Analytics, Exchange Program

Experience

Data Analyst, Deep ESG Sep 2024 – Nov 2024

- Developed and maintained Python notebooks to integrate ERP data into existing data pipelines
- *Technologies: Python, Pyspark, Pandas, PostgreSQL, Google Cloud Platform, Git, Github*

Backend Software Engineer, Sisloc Softwares March 2021 – Nov 2022

- Rewrote and optimized legacy SQL queries, enhancing system performance and reducing query execution time
- Maintained and refactored legacy code written in Delphi, diagnosing and fixing critical bugs to improve stability
- Designed and implemented new features, integrating them into the existing legacy system architecture
- *Technologies: SQL Server, Delphi, Agile Methodologies*

Projects

Identifying and Analyzing Communities in YouTube Comment Sections repository link

- An academic research project exploring the interaction patterns and community dynamics in YouTube comment sections, focusing on content creators targeting younger audiences.
- Developed an analysis system combining community detection with the Louvain Algorithm, NLP techniques including Topic Modeling and Sentiment Analysis, and clustering approaches for pattern identification.
- *Technologies: Python, NetworkX, NLTK, Pandas, Matplotlib, Scikit-Learn*

Semantic-Driven BERTopic: Comparative Analysis of Embedding Techniques repository link

- An academic project implementing a semantic-driven topic modeling pipeline combining Sentence Transformers with UMAP dimensionality reduction and HDBSCAN clustering, integrated with the OCTIS evaluation framework.
- Applied a comparative methodology across embedding techniques (SimCSE, GTE-multilingual, MPNet) with a coherence-driven topic merging step to refine cluster outputs.
- *Technologies: PyTorch, Sentence Transformers, SimCSE, UMAP, HDBSCAN, BERTopic, OCTIS, scikit-optimize*

Context-Aware Implicit Hate Speech Detection repository link

- A project translating implicit hate speech datasets (IHC, SBIC, DynaHate) from English to Portuguese through machine translation, with quality assessment via COMETKIWI and SBERT metrics
- Adapted and benchmarked the SHAREDCON contrastive learning method, which clusters implicit speech based on shared semantics, for Portuguese-language detection
- *Technologies: Sentence Transformers, SBERT, COMETKIWI, Contrastive Learning, Machine Translation, NLP*